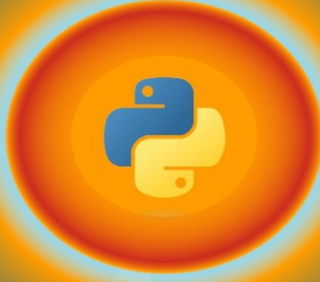


Sriphagaarucht Srinhichaarnun

Rachanee Kalayavinai



A-Z
IN
PYTHON

PROGRAMMER

FROM BEGINNER TO



CHAPTER 1 INTRODUCTION

Python is a popular programming language created by Guido van Rossum and launched in 1991. It can be used to build web applications, servers, in combination with other software to create workflows, connect to a relational database system that uses Structured Query Languages (SQLs) such as Oracle, DB2 or MySQL, as well as connect to non-relational database systems (NoSQL) such as MongoDB. This is explained in detail in the Advanced Python Programming Textbook.

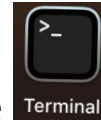
Advantages of the Python language include:

1. Being able to handle a lot of data and perform complex mathematical calculations.
2. Rapid prototyping or development of production-ready software
3. Works on different OSs (Windows, Mac, Linux, Raspberry Pi, etc.)
4. Simple grammar similar to English, making it easy to program.
5. It contains syntax that allows programmers to write programs with a smaller number of lines compared to other programming languages.
6. Works on interpreter systems, so codes can be executed as soon as it is written, which means prototyping can be done quickly.
7. Able to act in a procedural, object-oriented or function-oriented manner.
8. Python uses a line break to complete a command, as opposed to other programming languages that often use semicolons or

brackets. Relying on indents using spaces to define boundaries, such as loop boundaries, functions, and classes, other programming languages often use brackets for this purpose.

Installing Python

Many PCs and Macs have Python pre-installed. To check, launch the Terminal App on Mac, or Command Line on the PC and enter the command below.



For Mac or Linux OSs, go to the terminal, the console will appear to enter the command, with the host name appearing differently depending on the name defined. The author uses the name “VariitSris”, so the following line of command will be displayed.

```
VaritSris> █
```

To find out if the machine has Python installed and which version, use the following command:

```
VaritSris> python3 --version
```

```
Python 3.12.3
```

Note: For Windows operating systems, go to the Command Line and use the `py --version` command.

If not installed, go to:

<https://www.python.org/downloads>

To download the latest version of python according to your operating system, as shown in Figure 1-1: download window for the latest version of

python.